Supplementary Material
bioMoR: Biology-Guided Mixture-of-Recursions for Efficient and Interpretable Genomic Learning

Anonymous Submission

This supplement collects the additional tables and derivations referenced in the main paper. Unless stated otherwise, the auxiliary empirical tables here use a 3-seed held-out protocol; the 5-fold cross-validation results reported in the main paper are the authoritative numbers, and we include these analyses for completeness and transparency.

A Baron Training-Cost Curves

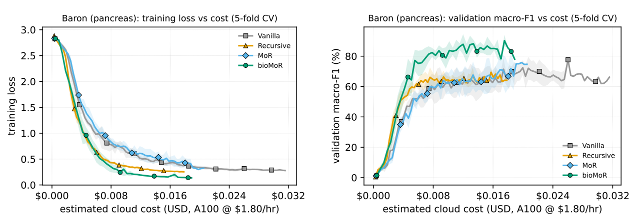

Figure 1: **Full training-cost curves on Baron (pancreas; 5-fold CV).** Training loss (left) and validation macro-F1 (right) vs. measured A100 GPU cost, for all four architectures; lines are the mean over the same 5 folds as the main table, shaded bands ± 1 SD. bioMoR reaches its accuracy plateau at lower cumulative training cost than the vanilla transformer. The main paper summarises these curves as an accuracy-vs-cost point diagram (best-loss and best-accuracy operating points).

B PATH-Comparable Positive-Class F1

C Model-Size Scaling

D Dataset Details

We use 14 datasets: 8 genomap single-cell datasets (Islam and Xing 2023), each an expression matrix with cell-type labels and a fixed stratified split – Baron, Lung, Muraro, Oesophagus, Segerstolpe, Spleen, T-cell and Xin – 3 Reactome multi-omics cohorts — prostate, bladder (BLCA), stomach (STAD) cancer — whose fixed Reactome pathway tokens pool mutation, copy-number and expression channels, and 3 pan-cancer TCGA settings of a binary primary-vs-metastatic task (PM, mutation+CNV; PC, expression; 3M, tri-modal).

Model	Prostate	Bladder	Stomach	PM	PC	3M
Vanilla	64.6$\pm$9.0	78.8$\pm$2.2	66.7$\pm$2.6	57.0$\pm$6.3	89.3$\pm$2.5	88.7$\pm$2.2
Recursive K=2	65.0$\pm$8.0	78.0$\pm$2.7	66.7$\pm$6.4	52.5$\pm$8.9	88.6$\pm$4.6	90.6$\pm$3.9
Recursive K=3	64.6$\pm$6.8	78.4$\pm$2.1	64.4$\pm$5.8	54.3$\pm$9.2	90.3$\pm$2.1	89.3$\pm$2.5
Recursive K=4	60.3$\pm$8.2	74.6$\pm$5.2	65.5$\pm$5.5	51.8$\pm$5.9	87.3$\pm$3.9	88.2$\pm$3.5
MoR-Expert K=2	65.8$\pm$8.6	80.3$\pm$1.2	66.3$\pm$4.0	57.5$\pm$3.8	88.1$\pm$4.8	88.7$\pm$3.6
MoR-Expert K=3	62.3$\pm$7.8	78.2$\pm$3.6	68.3$\pm$3.7	54.3$\pm$6.9	89.7$\pm$2.0	89.3$\pm$5.9
MoR-Expert K=4	63.6$\pm$8.9	77.4$\pm$3.6	67.0$\pm$3.9	52.9$\pm$5.2	88.9$\pm$3.0	90.6$\pm$3.7
Token-choice K=2	65.7$\pm$8.2	78.3$\pm$3.9	66.6$\pm$6.0	58.1$\pm$5.2	87.9$\pm$2.7	87.9$\pm$1.6
Token-choice K=3	71.2$\pm$5.2	76.9$\pm$3.9	65.3$\pm$8.1	52.6$\pm$7.9	88.0$\pm$4.5	89.3$\pm$3.2
Token-choice K=4	70.0$\pm$7.6	78.4$\pm$3.0	65.8$\pm$5.0	55.7$\pm$5.8	87.2$\pm$3.0	89.6$\pm$3.6

Table 1: **PATH-comparable positive-class F1 on the multi-omics cohorts.** Each cell is the threshold-calibrated positive-class (progression: metastatic / late-stage) F1, mean $\pm$ SD over the folds (%), for the full efficiency ladder — the positive-class counterpart of the macro-F1 main-results table (Table 2 of the main paper). Following the PATH protocol, the operating threshold is tuned on validation to maximise positive-class F1 and applied to the held-out test fold, with AUROC-based model selection; all other settings match the main-results table. PM, PC and 3M are the pan-cancer cohorts under mutation+CNV, expression, and tri-modal (all three) inputs; Bladder/Stomach are the BLCA/STAD Reactome cancers.

Per-dataset sample, feature and class counts are read directly from the result files and summarised in Table 8.

E Implementation Details and Hyperparameters

All settings below are held *fixed across every dataset and every variant* (no per-dataset tuning); the only quantities we vary are those explicitly swept in a reported study (marker/token budget M in the scaling table, recursion depth K in the ladder). This keeps every comparison in the main results paired.

Optimisation and protocol. We train with AdamW (learning rate 3×10^{-4}, weight decay 10^{-2}), dropout 0.1, for up to 100 epochs with early stopping on validation macro-F1 (patience 15). The protocol is 5-fold stratified cross-validation (seed 42; 20% test, 10%-of-train validation), with per-gene z-scoring fit on the training fold only. Single-cell models use d_{model}=96 and M=128 learnable marker tokens; multi-

d_{model}	Single-cell			Multi-omics (Reactome)		
	Vanilla	MoR-gen.	bioMoR	Vanilla	MoR-gen.	bioMoR
96	63.6±5.9	63.4±4.1	74.8±2.0	52.1±5.1	55.6±9.0	56.2±4.3
136	65.1±5.0	62.4±5.6	75.8±3.8	51.2±7.7	66.0±9.9	53.5±5.6
192	61.8±5.1	62.8±4.3	76.2±3.3	42.5±6.1	*run…*	54.8±4.2
272	60.7±5.8	63.5±4.6	75.4±3.5	52.8±4.8	33.5±6.4	57.0±2.4
352	59.7±5.6	61.4±6.5	77.1±4.0	48.6±8.0	47.3±4.8	56.1±4.5

Table 2: **Model-size scaling.** Macro-F1 (mean±SD) across a width sweep ($d_{\text{model}} \in \{96, \ldots, 352\}$) for Vanilla, MoR-general and bioMoR, averaged over the 8 single-cell and 3 Reactome multi-omics suites. bioMoR holds a flat, high accuracy across widths while the vanilla stack—at $4\times$ the recursion-stack parameters—degrades.

Dataset	M=128	M=256	M=512	M=1024	M=2048	Linear (all feat.)
Baron	79.1 ± 2.8	78.1 ± 1.4	79.6 ± 4.4	82.6 ± 2.0	80.4 ± 2.1	91.1 ± 2.0
Lung	79.1 ± 1.2	80.2 ± 0.8	79.8 ± 1.9	81.3 ± 2.0	81.1 ± 1.9	74.8 ± 0.2
Muraro	75.3 ± 6.4	72.4 ± 2.7	68.3 ± 3.3	75.7 ± 4.5	73.8 ± 6.0	95.7 ± 2.2
Oesophagus	69.6 ± 1.4	69.8 ± 0.6	71.5 ± 1.6	73.6 ± 0.5	72.2 ± 1.6	69.5 ± 1.1
Segerstolpe	71.4 ± 5.3	70.3 ± 4.1	73.8 ± 5.4	69.4 ± 14.7	68.3 ± 3.3	89.5 ± 5.0
Spleen	58.2 ± 0.9	58.6 ± 1.6	59.9 ± 0.8	61.1 ± 2.3	61.6 ± 4.0	64.7 ± 0.1
T-cell	68.9 ± 1.0	69.9 ± 0.3	71.5 ± 0.4	71.8 ± 2.3	73.0 ± 0.3	78.1 ± 0.3
Xin	64.7 ± 9.4	64.8 ± 6.0	63.7 ± 10.3	64.1 ± 0.6	62.1 ± 2.8	97.4 ± 0.6
Mean macro-F1	**70.8**	**70.5**	**71.0**	**72.5**	**71.6**	**82.6**

Table 3: **Marker-budget headroom (3-seed held-out).** Single-cell macro-F1 as the marker budget M grows, against the full-feature linear baseline. Even when every feature is a marker, a residual to the linear model remains, so the gap is architectural rather than a token-budget limitation.

omics models use d_{model}=128, M=256 fixed Reactome pathway tokens, and batch size 32. The weight-shared block is applied up to K=4 times by default.

Router budgets. Expert-choice routing uses a fixed *capacity funnel*: at recursion step t the router keeps a fraction $c_t = \max(0.75,\, 1 - 0.25\,t)$ of the tokens, i.e. $c = (1.00,\, 0.75,\, 0.75,\, 0.75)$ at K=4 (all tokens pass step 0, then a 0.75 floor thereafter; the floor is exposed as `RMT_CAP_FLOOR`). The floor is deliberately gentle—because the classifier mean-pools the M marker tokens, freezing too many too early starves the pooled vector, and a 0.5 floor made K=4 the weakest depth. Token-choice routing instead lets each token self-gate a single depth in $\{1, \ldots, K\}$ (top-1 over depths), so its budget is emergent rather than fixed. The router gate scale (α=1.0) and temperature (τ=1.0) are left at their neutral defaults.

Balancing coefficients. Expert-choice is load-balanced by construction (the funnel fixes per-step capacity), so it carries no balancing loss. Token-choice adds a Switch-style load-balancing term (Shazeer et al. 2017) over the K depths with weight 0.1, plus a router z-loss (logit-magnitude regulariser) with weight 10^{-3}. When the biological prior is injected at the router it enters as an additive centrality bias on the depth logits with initial strength β_0=1.0, annealed toward 0 over training; the learned interaction graph is mixed into the token embeddings through a sigmoid-bounded, *learnable* coefficient initialised at λ=0.3. A marker-sufficiency auxiliary head (weight 1.0) probes the pre-recursion marker panel; the

Dataset	Learned (random init)	+ bio warm-start	+ stronger init	+ annealed anchor
Baron	82.3 ± 2.2	80.9 ± 3.8	81.6 ± 1.8	66.8 ± 6.2
Lung	79.5 ± 1.4	79.8 ± 0.8	78.4 ± 0.6	70.0 ± 2.7
Muraro	74.5 ± 5.4	75.3 ± 4.9	76.6 ± 5.4	75.9 ± 9.3
Oesophagus	69.3 ± 0.7	71.1 ± 0.9	70.2 ± 1.0	57.4 ± 2.6
Segerstolpe	74.0 ± 5.3	69.9 ± 7.4	74.5 ± 5.7	53.0 ± 5.0
Spleen	59.0 ± 0.2	59.1 ± 0.7	58.7 ± 1.2	52.5 ± 1.2
T-cell	68.9 ± 0.7	69.1 ± 1.1	69.0 ± 0.6	54.7 ± 2.4
Xin	69.2 ± 5.0	67.2 ± 8.4	68.5 ± 6.1	69.5 ± 7.7
Mean macro-F1	**72.1**	**71.5**	**72.2**	**62.5**
Δ vs. random	+0.0	−0.6	+0.1	−9.6

Table 4: **Stress-testing the biological warm-start** (single-cell macro-F1, mean±std). From a randomly-initialised learned graph we add a biological warm-start, then a stronger init, then an annealed anchor $\lambda(t)\|A_{\text{learned}} - A_{\text{bio}}\|_F^2$ that keeps the graph near biology early. The Δ row is the mean gain over random init; forcing biology in more strongly does not beat learning the graph from data.

Cohort	bioMoR (mut+CNV)	Geneformer	scGPT
Prostate	78.2 ± 2.2	78.2 ± 3.3	40.1 ± 0.0
BLCA (bladder)	40.9 ± 1.9	47.7 ± 3.7	40.4 ± 0.0
STAD (stomach)	52.2 ± 6.8	46.1 ± 5.5	35.7 ± 0.0
Mean	**57.1**	**57.3**	**38.7**

Table 5: **bioMoR vs. gene-vocabulary foundation models** on the Reactome multi-omics cohorts (macro-F1, mean±std over seeds). Geneformer (fine-tuned) and scGPT (frozen embedding + logistic probe) are mapped onto each cohort's HUGO gene symbols with a per-gene mutation/copy-number alteration burden, on the same splits as bioMoR. Bulk DNA-alteration input is out-of-distribution for these single-cell-RNA models.

total objective is the classification cross-entropy plus these auxiliary and balancing terms.

F Effective-FLOPs Accounting

The compute numbers report per-sample FLOPs of the recursive transformer stack, the only component routing changes. One application of the shared block to a tokens costs $\phi(a) = 4a^2 d + 4\,a\,d\,d_{\text{ff}}$; the nominal fixed-depth cost is $\Phi_{\text{nom}} = K\,\phi(M)$ and the effective cost sums one block over the tokens active at each step, $\Phi_{\text{eff}} = \sum_{t=1}^{K} \phi(a_t)$, where a_t is the mean number of markers the expert-choice funnel keeps at step t. Gene embedding and marker selection are $\mathcal{O}(Nd)$, identical across routing modes, and excluded from this stack-level comparison.

End-to-end accounting. For completeness we account for the whole forward pass, not only the stack. Three parts scale with the full gene count N: gene embedding $\Theta(Nd)$; the cross-attention marker router, whose M queries attend over all N gene keys at $\Theta(MNd)$; and the optional gene-graph smoother, which at rank r costs $\Theta(Nr)$ (never the N^2 dense form). Everything after marker selection – the recursive stack, pooling and head – scales with the compressed budget $M \ll N$ at $\Theta(M^2 d)$ per pass. So the router's $\Theta(MNd)$ term dominates the N-scaling and is *linear* in N (versus the $\Theta(N^2 d)$ self-attention a full-gene transformer would pay),

Depth K	Shared (ours)	Independent	Reduction
1	74,880	74,880	$1\times$
2	74,880	149,760	$2\times$
3	74,880	224,640	$3\times$
4	74,880	299,520	$4\times$
6	74,880	449,280	$6\times$
8	74,880	599,040	$8\times$

Table 6: **Parameter reduction.** One shared block versus K independent layers at matched width: an exact $K\times$ reduction, present before any training.

Dataset	M=16	M=32	M=64	M=128	M=256
Single-cell (genomap)					
Baron	33.7 ± 7.1	38.5 ± 2.1	53.9 ± 1.9	62.6 ± 5.1	66.9 ± 3.4
Lung	30.8 ± 5.9	38.1 ± 1.7	61.5 ± 1.9	71.4 ± 1.1	78.4 ± 0.8
Muraro	55.8 ± 3.6	59.4 ± 1.3	67.9 ± 5.6	68.5 ± 4.1	76.2 ± 5.2
Oeso.	23.1 ± 1.3	30.2 ± 3.9	42.0 ± 2.4	53.9 ± 4.4	63.2 ± 2.4
Seger.	41.2 ± 12.1	45.3 ± 5.6	59.3 ± 4.0	59.1 ± 1.6	55.4 ± 7.0
Spleen	18.7 ± 2.4	26.8 ± 3.0	38.1 ± 2.3	48.1 ± 5.0	59.4 ± 0.9
T-cell	25.5 ± 3.2	36.6 ± 5.4	43.1 ± 5.2	47.4 ± 2.2	61.5 ± 1.7
Xin	50.5 ± 5.9	46.2 ± 1.6	60.8 ± 5.6	64.1 ± 3.3	72.4 ± 11.7
Multi-omics (Reactome)					
Prostate	74.3 ± 3.0	74.7 ± 2.7	78.0 ± 3.0	76.7 ± 2.6	77.3 ± 0.2
BLCA (bladder)	49.8 ± 2.2	46.0 ± 3.1	49.3 ± 2.5	48.1 ± 6.8	55.7 ± 3.5
STAD (stomach)	54.2 ± 3.1	51.0 ± 2.2	44.0 ± 5.6	51.2 ± 2.8	49.4 ± 4.1
Mean macro-F1	**41.6**	**44.8**	**54.4**	**59.2**	**65.1**

Table 7: **Marker-token budget.** Macro-F1 (mean$\pm$std over 3 seeds) as the marker budget M shrinks from 256 to 16. A few dozen to a few hundred tokens recover most of the full-gene accuracy. Bottom row: mean over the 11 single-cell and Reactome multi-omics datasets included in this sweep (the pan-cancer settings are omitted here).

while the quadratic cost is paid only on the M markers. The architecture ablations of the main-paper efficiency ladder vary only the stack, so the stack-level ratios there are the correct comparison for the routing/sharing claims; the router and embedding terms are shared by every variant. Measured wall-clock and peak-memory profiling on matched hardware is a straightforward addition we leave to the camera-ready.

G Theoretical Foundation of the Router

This appendix gives the mathematical and biological grounding for bioMoR's biology-informed router.

Routing as conditional computation. The router implements *conditional computation*: a learned policy routes each token to a token-specific amount of compute, the gating principle of sparsely-gated mixtures of experts (Shazeer et al. 2017), reused by Mixture-of-Depths (Raposo et al. 2024) and Mixture-of-Recursions (Bae et al. 2025); the "experts" here are recursion *depths* of one shared block, which couples adaptive computation (Graves 2016) to weight sharing.

The differentiable handle. The discrete top-k has zero gradient almost everywhere, so bioMoR keeps the soft probability of the chosen route as a multiplicative *gate* on the block output, $\mathbf{o}_m = g_m\, f_\theta(\mathbf{h}_m)$ with $g_m = \sigma(\tilde{r}_m)$. Because g_m is smooth in $\mathbf{w}_r$, the chain $\mathcal{L} \leftarrow \mathbf{o}_m \leftarrow g_m \leftarrow \mathbf{w}_r$ is unbroken:

Dataset	Modality	Samples	Features	Classes
Baron	single-cell	8,569	2,000	13
Lung	single-cell	57,020	1,089	25
Muraro	single-cell	2,122	2,000	9
Oesophagus	single-cell	87,947	1,089	18
Segerstolpe	single-cell	2,127	2,000	11
Spleen	single-cell	94,129	1,089	28
T-cell	single-cell	24,007	1,089	7
Xin	single-cell	1,492	2,000	4
Prostate	multi-omics	1,011	1,223	2
BLCA (bladder)	multi-omics	404	1,258	2
STAD (stomach)	multi-omics	414	1,258	2
Pan-cancer PM (mut+CNV)	multi-omics	8,893	10,333	2
Pan-cancer PC (expression)	multi-omics	8,586	10,013	2
Pan-cancer 3M (tri-modal)	multi-omics	8,586	9,880	2

Table 8: The datasets used throughout (14 in total): 8 genomap single-cell suites, 3 Reactome multi-omics cohorts, and 3 pan-cancer TCGA settings (PM, PC, 3M). Counts are read directly from the result files.

the hard choice routes, the soft gate carries the gradient. The biological prior of the routing logit of the main paper (Eq. 1) is an additive constant in this logit, so it shifts the decision without breaking this path.

Biological foundation of the prior. Two facts from single-cell biology motivate the prior. A small set of *marker* genes carries most discriminative signal (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023), so compute should be allocated unevenly. And gene co-expression networks are approximately scale-free: a few high-degree *hub* genes (master regulators) exert outsized influence. We operationalise "hub" as eigenvector centrality on the genomap co-expression graph $\mathbf{W}$: the leading eigenvector $\mathbf{v}$ of $\mathbf{W}\mathbf{v} = \lambda\mathbf{v}$ scores each gene by the centrality of its neighbours, recursively, and we z-score it to form π.

Empirical-Bayes reading. The routing logit of the main paper (Eq. 1) is a log-linear prior on the routing decision, with $\beta_t\, \pi_m$ a Gaussian-like prior mean and $\beta_t = \beta_0(1-\text{progress})$ a shrinkage strength that decays as data evidence accumulates: prior-dominated when the likelihood is uninformative (early training), data-dominated later.

Leakage safety. $\mathbf{W}$ is computed from training-split *expression only*; no cell-type label enters it. The prior therefore injects network topology, not the answer, which is why a gene-discovery claim under this prior is not circular, unlike a prior built from curated cell-type marker lists.

Optional pathway-graph smoothing. The additive bias treats genes independently. Co-pathway genes should route coherently, which one obtains by smoothing the logits over $\mathbf{W}$ with the normalised Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, $\hat{\mathbf{r}}^{(t)} = \tilde{\mathbf{r}}^{(t)} - \gamma\, \mathbf{L}\tilde{\mathbf{r}}^{(t)}$ (graph-Laplacian regularisation): a gene borrows routing strength from its network neighbours. This is one sparse matrix-vector product per step and adds no transformer parameters; we expose it as an option and leave its evaluation to future work.

H Routing-Depth Visualisation

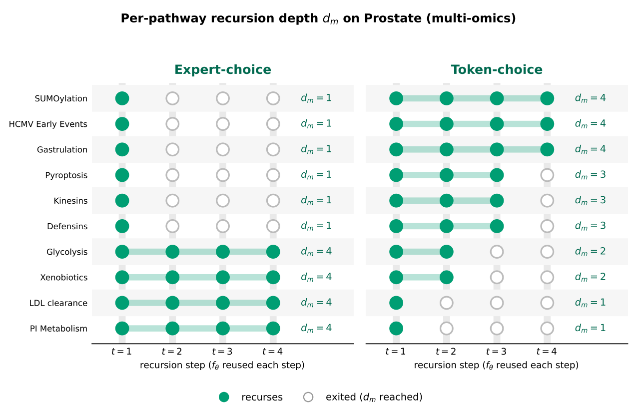

Figure 2: **Two routing modes assign per-pathway depth** d_m (real Prostate tokens; • recurses, ◦ exited; the block f_θ is reused up to K=4, so depth adds no parameters). Expert-choice fixes capacity per step; token-choice lets each pathway self-gate its depth.

I Per-Pathway Recursion-Depth Tables

Complete keep/drop rankings for the PM cohort (pan-cancer, mutation+copy-number; 1258 Reactome pathway tokens, 8893 samples) under canonical bioMoR at K=4, referenced from the main-text Interpretability section. Mean recursion depth $d_m \in [0, 4]$ is averaged over the same 5 held-out folds as the main table. Higher d_m = the router keeps allocating compute (*kept*); the minimum = early exit (*dropped*). The two routing rules keep *different* pathways: their top-25 kept sets share 0 pathways and their per-pathway depths are weakly anti-correlated (Spearman −0.28), so each ranking is read within its routing mode.

Expert-choice (K=4, Macro-F1 83.9±1.9)

Kept — 25 deepest-routed			Dropped — 25 earliest-exit		
#	Pathway	d_m	#	Pathway	d_m
1	R-HSA-1632852	4.00	1	R-HSA-977606	1.73
2	R-HSA-2028269	3.99	2	R-HSA-166663	1.74
3	R-HSA-74751	3.99	3	R-HSA-166658	1.81
4	R-HSA-9663891	3.99	4	R-HSA-202433	1.81
5	R-HSA-2565942	3.99	5	R-HSA-2029481	1.84
6	R-HSA-9678108	3.99	6	R-HSA-391160	1.94
7	R-HSA-9706574	3.99	7	R-HSA-6803529	1.97
8	R-HSA-2404192	3.99	8	R-HSA-1483206	1.99
9	R-HSA-442982	3.99	9	R-HSA-3371568	2.01
10	R-HSA-8963899	3.99	10	R-HSA-9664323	2.05
11	R-HSA-918233	3.99	11	R-HSA-111465	2.19
12	R-HSA-5628897	3.99	12	R-HSA-9828806	2.20
13	R-HSA-3769402	3.98	13	R-HSA-9830364	2.21
14	R-HSA-9013418	3.98	14	R-HSA-4419969	2.21
15	R-HSA-9833110	3.98	15	R-HSA-202131	2.23
16	R-HSA-445989	3.98	16	R-HSA-1483166	2.24
17	R-HSA-499943	3.98	17	R-HSA-9820965	2.24
18	R-HSA-2559580	3.98	18	R-HSA-2173788	2.24
19	R-HSA-168898	3.98	19	R-HSA-171319	2.24
20	R-HSA-109704	3.98	20	R-HSA-606279	2.25
21	R-HSA-9818027	3.98	21	R-HSA-1482925	2.25
22	R-HSA-442742	3.98	22	R-HSA-8862803	2.25
23	R-HSA-5693571	3.98	23	R-HSA-76071	2.25
24	R-HSA-1566977	3.98	24	R-HSA-212300	2.25
25	R-HSA-69206	3.98	25	R-HSA-352230	2.26

Token-choice (K=4, Macro-F1 85.3±3.6)

Kept — 25 deepest-routed			Dropped — 25 earliest-exit		
#	Pathway	d_m	#	Pathway	d_m
1	R-HSA-982772	3.82	1	R-HSA-170834	1.28
2	R-HSA-9828806	3.82	2	R-HSA-68949	1.36
3	R-HSA-9836573	3.80	3	R-HSA-2565942	1.41
4	R-HSA-977068	3.76	4	R-HSA-373076	1.42
5	R-HSA-9820965	3.75	5	R-HSA-9006936	1.42
6	R-HSA-9970672	3.74	6	R-HSA-5673001	1.42
7	R-HSA-9821002	3.71	7	R-HSA-5655302	1.44
8	R-HSA-9931295	3.71	8	R-HSA-909733	1.44
9	R-HSA-5334118	3.70	9	R-HSA-9637690	1.44
10	R-HSA-937072	3.66	10	R-HSA-69052	1.45
11	R-HSA-8931838	3.66	11	R-HSA-1643713	1.46
12	R-HSA-9710421	3.65	12	R-HSA-8878159	1.47
13	R-HSA-196807	3.65	13	R-HSA-2173793	1.49
14	R-HSA-199220	3.64	14	R-HSA-5633008	1.49
15	R-HSA-3238698	3.64	15	R-HSA-5668541	1.50
16	R-HSA-9821993	3.61	16	R-HSA-2559586	1.50
17	R-HSA-187687	3.61	17	R-HSA-9948299	1.52
18	R-HSA-1810476	3.61	18	R-HSA-418038	1.54
19	R-HSA-9619483	3.60	19	R-HSA-380320	1.55
20	R-HSA-9755779	3.58	20	R-HSA-9820952	1.55
21	R-HSA-5635838	3.58	21	R-HSA-9680350	1.56
22	R-HSA-1912420	3.58	22	R-HSA-69306	1.57
23	R-HSA-445144	3.58	23	R-HSA-8864260	1.58
24	R-HSA-9037629	3.57	24	R-HSA-1483257	1.58
25	R-HSA-70221	3.57	25	R-HSA-5683057	1.58

J Proofs for the Theoretical Analysis

We use the signal-plus-noise model of the main text: $x = z + \varepsilon$, $\varepsilon \sim (0, \sigma^2 I_G)$, $\varepsilon \perp z$, $\mathbb{E}z = 0$, $\Sigma = \mathbb{E}[zz^\top] =$

$\sum_k s_k u_k u_k^\top$ with $s_1 \geq s_2 \geq \cdots \geq 0$ and orthonormal $\{u_k\}$; at least one $s_k > 0$. The propagation is the linear filter $T_\lambda = (1-\lambda)I + \lambda A$, $A = \tilde{E}\tilde{E}^\top \succeq 0$, $\mathrm{rank}(A) \leq r$, and $R(T) = \mathbb{E}\|Tx - z\|^2$. Regimes: B_0 ($T{=}I$) $\subseteq B_1$ ($A{=}0$) $\subseteq B_2$ (learned A).

Lemma 1 (Spectral decoupling). For symmetric $T = \sum_k f_k u_k u_k^\top$, $R(T) = \sum_k [(1-f_k)^2 s_k + f_k^2 \sigma^2]$. *Proof.* $R(T) = \mathbb{E}\|(T-I)z\|^2 + \mathbb{E}\|T\varepsilon\|^2$ (cross term vanishes as $\varepsilon \perp z$). By trace cyclicity the first is $\mathrm{tr}((T-I)\Sigma(T-I)^\top) = \sum_k (f_k - 1)^2 s_k$ and the second is $\sigma^2 \mathrm{tr}(TT^\top) = \sigma^2 \sum_k f_k^2$. $\square$

Lemma 2 (Per-direction Wiener optimum). Each summand is strictly convex in f_k and minimised at $f_k^\star = s_k/(s_k + \sigma^2) \in [0,1)$ with residual $s_k\sigma^2/(s_k+\sigma^2)$. *Proof.* $g(f) = (1-f)^2 s_k + f^2\sigma^2$ has $g'(f) = -2(1-f)s_k + 2f\sigma^2$, vanishing at $f_k^\star$; $g'' = 2(s_k+\sigma^2) > 0$. Substituting gives the residual. The signal-preservation gain over the fully-shrunk value $g(0) = s_k$ is $s_k - \frac{s_k\sigma^2}{s_k+\sigma^2} = \frac{s_k^2}{s_k+\sigma^2}$. $\square$

Lemma 3 (Realizability). For any $\{f_i\}_{i\leq r} \subset [0,1]$ on any orthonormal $\{v_i\}_{i\leq r}$ there exist $\lambda \in [0,1]$ and PSD $A = \tilde{E}\tilde{E}^\top$ of rank $\leq r$ with $T_\lambda v_i = f_i v_i$. *Proof.* Take $\lambda = 1$, $\mu_i = (f_i - (1-\lambda))/\lambda = f_i \geq 0$, $A = \sum_{i\leq r} \mu_i v_i v_i^\top$, $\tilde{E} = [\sqrt{\mu_1}v_1, \ldots, \sqrt{\mu_r}v_r]$; on $\mathrm{span}\{v_i\}^\perp$, $A{=}0$ so $T_\lambda = (1-\lambda)I$ only further shrinks noise directions. $\square$

Theorem 1. Monotonicity is immediate from the nesting $B_0 \subseteq B_1 \subseteq B_2$ (min of the same objective over a larger set). $R^\star(B_0) = G\sigma^2$ ($f_k \equiv 1$ in Lemma 1). By Lemma 3 place $v_i{=}u_i$, $f_i{=}f_i^\star$ for $i \leq r$ and $f_k{=}1-\lambda$ on the tail; Lemma 2 gives, per top-r direction, a saving $\sigma^2 - \frac{s_i\sigma^2}{s_i+\sigma^2} = \frac{\sigma^4}{s_i+\sigma^2}$, equivalently a signal-preservation gain $\frac{s_i^2}{s_i+\sigma^2}$, and $\lambda \to 0$ recovers σ^2 on the tail. Hence $R^\star(B_0) - R^\star(B_2) \geq \sum_{i\leq r} \frac{s_i^2}{s_i+\sigma^2} > 0$. $\square$

Proposition 1 (No graph cannot close the gap). If $s_i \neq s_j$ for some $i,j \leq r$ then $R^\star(B_1) > R^\star(B_2)$. *Proof.* B_1 forces one scalar $f_k \equiv f$ (since $A{=}0$); minimising Lemma 1 over a single f gives $\hat{f} = \mathrm{tr}\Sigma/(\mathrm{tr}\Sigma + G\sigma^2)$, whereas the per-direction optima $f_i^\star$ differ when the s_i differ. A convex objective under the equality constraint "all f_k equal" exceeds its unconstrained minimum whenever the latter violates the constraint. $\square$

Proposition 2 (Task-risk bound). If the head Φ is L_Φ-Lipschitz and the label depends on the clean signal through the selection statistics Wz, then $\mathbb{E}\|\Phi(WTx) - \Phi(Wz)\| \leq L_\Phi \|W\|_2 \sqrt{R(T)}$; minimising R over B_2 strictly tightens this over B_0 by Theorem 1. *Proof.* Lipschitzness and the operator-norm inequality bound the LHS by $L_\Phi\|W\|_2\|Tx - z\|$; take expectations and apply Jensen. $\square$

Theorem 2 (Router monotone-safety). With router logit $\ell_t = \rho_\theta^{(t)}(H) + \Phi_t(AH)^\top$ and A row-stochastic, setting $\Phi{=}0$ recovers the baseline router exactly, so $\Theta_0 \subset \Theta_G$ and $R^\star(\Theta_G) \leq R^\star(\Theta_0)$; gradient flow from $\Phi{=}0$ has $\frac{d}{d\tau}L = -\|\nabla L\|^2 \leq 0$, so the trained router attains loss $\leq$ the baseline. Strictness: a baseline router is $\sigma(h_m)$-measurable and cannot separate two tokens with equal h_m but different neighbourhoods $(AH)_m$, incurring a fixed excess $\delta > 0$ that a suitable Φ removes. $\square$

Corollary 1 (Both sites best). The joint family contains embedding-only ($\Phi{=}0$) and router-only ($\lambda{=}0$) as no-op restrictions, so $R^\star_{\text{both}} \leq \min\{R^\star_{\text{emb}}, R^\star_{\text{rt}}\} \leq R^\star_{\text{none}}$ by monotonicity of the min over set inclusion. $\square$

K Low-Rank Propagation, Over-Smoothing, and Rare Cells

The denoising filter and the over-smoothing failure mode are two ends of the same spectral dial. Let $A = \tilde{E}\tilde{E}^\top$ have eigendecomposition $A = \sum_k a_k w_k w_k^\top$ with $a_k \geq 0$ and $a_k{=}0$ for $k > r$ (the range of $\tilde{E}$ is r-dimensional). Then $T_\lambda = (1-\lambda)I + \lambda A$ is simultaneously diagonalised, with eigenvalues

$$\mu_k(\lambda) \;=\; 1 - \lambda + \lambda a_k,$$

so on the r graph directions T_λ scales by $1 - \lambda + \lambda a_k$ and on the orthogonal complement it acts as the scalar $1-\lambda$.

Spectral view of over-smoothing. As $\lambda \to 1$ the off-range eigenvalue $1 - \lambda \to 0$: every direction outside $\mathrm{span}(\tilde{E})$ is annihilated and token features collapse onto the r-dimensional graph subspace, the discrete analogue of graph over-smoothing (repeated propagation driving representations toward a low-rank fixed point). The *spectral gap* that controls how fast this happens is $\min_k \mu_k - (1-\lambda) = \lambda\, a^+_{\min}$ on the range versus $1-\lambda$ off it; a large λ widens the gap and accelerates the collapse. Because bioMoR *learns* λ through a sigmoid (initialised at 0.3; App. E) rather than fixing it at the denoising-maximal value, and because the residual identity path guarantees $\mu_k \geq 1 - \lambda > 0$ for all k, no direction is fully annihilated: the operator is a *bounded* smoother, not a projector.

Rare-cell implication. Rare populations are precisely the directions this dial threatens. Their signal typically lives in low-variance modes (s_k small, and frequently outside the dominant top-r subspace), where the Wiener-optimal gain $f_k^\star = s_k/(s_k+\sigma^2)$ from Lemma 2 is already small; additional smoothing ($\mu_k \to$ off-range $1-\lambda$) drives the effective gain toward zero and erases the very directions that separate a rare type from the bulk. The learned bounded λ is therefore not a convenience but the mechanism that protects rare-cell discriminability while still denoising the high-variance modes ($s_k \gg \sigma^2$, where $f_k^\star \to 1$ and smoothing is safe). A second structural safeguard is that propagation acts on the $M \ll N$ marker/pathway *tokens*, not on cells: it mixes feature channels within a token but cannot average two cells of different type into one, so it cannot induce the cell-level homogenisation that over-smoothing causes in cell–cell-graph GNNs.

Prediction and diagnostics. This analysis predicts a *graded* benefit: the learned graph should help most where the raw per-feature signal is weak (denoising dominates) and little where classes are already separable (smoothing is redundant and risks the rare-cell modes) — exactly the pattern in the main-results table (largest gains on the hard single-cell suites and the multi-omics cohorts, negligible where a task is near-saturated). Two direct falsifiers we leave to future work: a per-class recall breakdown restricted to the rarest populations (over-smoothing predicts the learned graph should never *lower* rare-class recall relative to no graph, given the bounded λ), and a sweep of λ against the induced spectral gap $\lambda a^{+}_{\min}$ measuring where the denoising/over-smoothing trade-off turns.

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